BACKEND MODULE DESIGN DOCUMENT

PART-4: LOGGING, TESTING, OPERATIONS & FINAL REVIEW

School ERP

Java 21 + Spring Boot 3 | Spring Data JPA + Hibernate | PostgreSQL + Redis

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Preamble

This document is Part-4, the final part, of the Backend Module Design Document (BMDD) v1.0 for the School ERP System, continuing directly from Parts 1-3. It defines backend-implementation-specific logging, audit, testing, code quality, deployment-readiness, monitoring, exception-reporting, production-support, and governance standards, closing the documentation chain from RGD v2.0 through to the operational backend codebase itself.

This document reuses RGD v2.0, Appendix A-N, the UI/UX Screen Specifications, MDB Parts 1-4, DDD Parts 1-4, ASD Parts 1-4, and BMDD Parts 1-3 by reference only, and does not regenerate any of them. It contains no source code and no implementation — every standard is stated as a rule only.

31. Logging Standards

Fixes backend-implementation-specific application of the Logging Standards already established in MDB Part-2 Section 20 and ASD Part-2 Section 19, mapped onto the layer structure fixed in BMDD Parts 1-2.

- Structured (not free-text) logging is used at every layer, with the request identifier (ASD Part-1 Section 9) propagated from Controller through Service to Repository, so every log line from a single request can be correlated together regardless of which layer emitted it.
- Log level follows the fixed convention already established in MDB Part-2 Section 20 (ERROR for unhandled/system exceptions, WARN for Business Rule violations and authorization failures, INFO for successful state-changing operations, DEBUG for non-production diagnostics) — a Service method never logs a routine, successful Business Rule pass at WARN or higher.
- Sensitive/PII field values (Appendix-G/H Sensitive Data Matrix) are never included in a log line at any level, in any layer — this rule applies uniformly to Controller, Service, Repository, and Mapper logging alike, with no per-module exception.
- Centralized log aggregation (MDB Part-4 Section 21, DDD Part-4 Section 36) remains the only supported log destination — a module never writes to a separate, module-specific log file or destination outside this aggregation pipeline.

32. Audit Standards

Fixes the precise backend-implementation placement of the mandatory AuditLogService call already established in MDB Part-2 Section 20, BMDD Part-2 Section 13, and DDD Part-3 Section 24.

- Every Service method that creates, updates, deletes (soft-deletes), approves, rejects, or otherwise changes persisted state calls AuditLogService as the second-to-last step of its fixed sequence (BMDD Part-2 Section 13), immediately before any domain event is raised — a state-changing method is never considered complete without this call having executed successfully.
- A read-only Service method (Search, Report generation) does not call AuditLogService for the read itself, but any resulting Export or Print action (per the relevant UI/UX Screen Specification's Buttons & Actions) is itself logged as a distinct auditable action, consistent with Appendix-J Section 16.
- AuditLog old_value/new_value snapshots (DDD Part-3 Section 24) are populated by the same Service method making the change, using the pre- and post-change entity state already available to it — a separate reconciliation process is never relied upon to reconstruct what changed after the fact.
- A failure to write to AuditLog is treated as a failure of the entire state-changing operation (Section 21, BMDD Part-3) — the underlying business change and its audit record commit or roll back together, never independently.

33. Backend Testing Standards

Fixes backend-implementation-specific application of the Test Levels already established in Appendix-M Sections 8-9 and referenced in MDB Part-3 Section 29 and ASD Part-4 Section 36.

- Every Service method enforcing a Business Rule (BMDD Part-2 Section 18) has at least one Unit Test per distinct rule outcome (pass and each documented failure/edge case), using mocked Repository and cross-module Service dependencies, consistent with Appendix-M Section 8's Unit Testing level.
- Repository methods are tested against a real (not mocked) database instance in the Testing environment (Appendix-N Section 5), verifying actual query behavior, pagination, and constraint enforcement (DDD Part-3) rather than relying solely on mocked Repository behavior in Service-layer tests.
- Integration Testing (Appendix-M Section 8) verifies the full layer flow already fixed in BMDD Part-2 Section 19 end-to-end for each API operation, including that the correct response category (ASD Part-1 Section 10) is returned for both success and every documented failure category.
- Test data is synthetic and isolated per test run (Appendix-M Section 12) — a backend test never depends on a specific pre-existing row surviving from a prior test run, avoiding order-dependent test fragility.

34. Code Quality Standards

Consolidates the backend coding and review standards already established in MDB Part-1 Section 10 and MDB Part-3 Sections 29-30 into a single operational reference.

- Linting/static analysis (MDB Part-1 Section 10.3) runs as a required build step (MDB Part-4 Section 34) — a build with lint or static-analysis findings above the agreed severity threshold does not produce a mergeable artifact.
- Every public Service and Repository method carries a documentation comment referencing the FR/BR/NFR/Entity ID it implements (MDB Part-3 Section 29), extending the traceability chain into the code itself, not only into commit messages and pull-request descriptions.
- Method and class complexity remain bounded by the same granularity principle already fixed in BMDD Part-2 Section 13 — a Service method handling multiple, loosely-related Business Rules is split into multiple, focused methods rather than grown into a single large method.
- No dead code, commented-out blocks, or unused imports/dependencies are merged (MDB Part-3 Section 29) — the codebase reflects only the currently approved specification set at all times.

35. Deployment Readiness

Fixes the backend-specific subset of the Production Readiness Checklist already established in MDB Part-4 Section 40 and DDD Part-4 Section 40, verified before any backend release.

- Every new/changed API operation has passed the API Review and API Security Review checklists already fixed in ASD Part-3 Section 30 and ASD Part-2 Section 20, in addition to the Backend Quality and Backend Performance checklists already fixed in BMDD Part-2 Section 20 and BMDD Part-3 Section 30.
- Database migrations accompanying the release (MDB Part-3 Section 28, DDD Part-4 Section 34) have been applied and verified in Staging against production-representative data volume before the corresponding backend release is promoted.
- Health check endpoints (MDB Part-4 Section 37) correctly reflect the readiness of any new dependency introduced by the release (a new external integration adapter, Section 26 BMDD Part-3) before that release reaches Production.
- Rollback viability (MDB Part-4 Section 36, Appendix-L Section 21) is confirmed for the specific release — a backend release is not promoted to Production without a validated path back to the prior working version.

36. Operational Monitoring

Fixes backend-implementation-specific monitoring signals, extending MDB Part-4 Section 37 and ASD Part-4 Section 34 with module-level detail.

- Every module's Service layer exposes operation-level metrics (invocation count, latency, error rate per method or method-group) feeding the same centralized monitoring pipeline already fixed in MDB Part-4 Section 37 — a module never maintains a separate, isolated metrics destination.
- Background Job (BMDD Part-3 Section 25) execution status (last-run time, success/failure, duration) is exposed as a monitored signal, so a silently-failing scheduled job is detected by monitoring rather than discovered only when its absence causes a downstream business impact.
- Cache hit/miss ratio (BMDD Part-3 Section 22) is monitored per cache-eligible entity type, informing whether the TTL and invalidation strategy (Sections 22-23, BMDD Part-3) remain well-tuned as usage patterns evolve.
- External integration adapter health (BMDD Part-3 Section 26) is monitored independently per integration, consistent with the per-channel isolation already fixed in Appendix-F NFR-MON-002.

37. Exception Reporting

Extends the centralized Exception Handling already fixed in BMDD Part-2 Section 16 with the reporting/escalation behavior for unhandled and repeated failures.

- Every Unhandled/System Exception (BMDD Part-2 Section 16) is reported to the centralized monitoring/alerting pipeline (MDB Part-4 Section 37) in addition to being logged (Section 31 this document) — an unhandled exception is never left visible only in a log file awaiting manual discovery.
- A Business Rule Exception or Validation Exception occurring at unusually high frequency for a specific operation (relative to its historical baseline) is itself surfaced as a monitoring signal, since a spike in expected-category errors often indicates an upstream defect (e.g., a frontend sending malformed requests) rather than genuine, isolated business-rule conflicts.
- Exception reports include the request identifier (ASD Part-1 Section 9) and the Appendix-C v1.1 rule ID where applicable (BMDD Part-2 Section 18), enabling rapid correlation between a monitoring alert and the specific operation/rule involved, without needing to first search application logs.

38. Production Support Standards

Fixes backend-specific production-support practice, extending MDB Part-4 Section 39 and DDD Part-4 Section 39's Change Management and Separation of Duties principles.

- A Production incident traced to a specific backend defect is remediated preferentially via rollback to the prior working release (MDB Part-4 Section 36) rather than an emergency hotfix, unless rollback itself is infeasible for the specific incident (e.g., an incident caused by a data condition that rollback alone would not resolve).
- An emergency hotfix, where used, follows an abbreviated but still-real version of the standard Code Review and Backend Quality checklists (BMDD Part-2 Section 20) — abbreviated review is faster, not absent, and is followed by full retroactive review within a defined window, per MDB Part-4 Section 39.
- Direct, manual data correction in Production (bypassing the standard Service-layer Business Rule enforcement, BMDD Part-2 Section 18) is an exceptional, logged, IT-Admin-approved action, never a routine support technique — a recurring need for manual correction is treated as a signal that the underlying Business Rule or defect needs a proper code fix, not a maintained workaround.

39. Backend Governance

Fixes ownership and decision authority for the standards fixed across all four BMDD Parts, mirroring the governance model already established for the ASD in ASD Part-4 Section 39.

- The Backend Architecture Team is the designated owner of this BMDD and all four of its Parts — any deviation from a standard fixed here requires that team's explicit, documented approval, not a unilateral per-module decision.
- This BMDD is reviewed for continued accuracy at the same cadence as the MDB, DDD, and ASD documents it depends on — a standard that has silently drifted from actual implementation practice is treated as a governance failure to be corrected, not tolerated as de facto reality.
- Any conflict between this BMDD and an earlier document in the chain (RGD v2.0 through ASD Part-4) is resolved in favor of the earlier, more foundational document — this BMDD elaborates and applies those decisions at the code-structure level, it does not have authority to silently override them.

40. Final Backend Checklist

The final consolidation point of the entire Backend Module Design Document (Parts 1-4), verified before any major release alongside the Production Readiness Checklist (MDB Part-4 Section 40), Database Operations Checklist (DDD Part-4 Section 40), and API Final Review Checklist (ASD Part-4 Section 40).

#	Checklist Item
1	Module and package structure (BMDD Part-1 Sections 3-4) intact for every new/changed component
2	Layer responsibilities and dependency rules (BMDD Part-1 Sections 5-6) respected with no violation
3	Entity, Repository, Service, and Controller standards (BMDD Part-2 Sections 11-14) followed exactly
4	Validation, exception handling, transaction, and Business Rule enforcement (BMDD Part-2 Sections 15-18) correct for every changed operation
5	Transaction propagation, caching, events, background jobs, integrations, file processing, and concurrency (BMDD Part-3 Sections 21-28) correctly implemented
6	Logging and Audit (Sections 31-32 this document) present and correctly placed for every state-changing operation
7	Testing coverage (Section 33) meets Appendix-M's Unit and Integration levels for every changed Business Rule
8	Code quality (Section 34) passes linting/static analysis with full FR/BR/Entity traceability in documentation comments
9	Deployment readiness and monitoring (Sections 35-36) confirmed before Production promotion
10	No undocumented deviation from BMDD Parts 1-4 exists without a logged governance exception (Section 39)

A negative finding against any item above is treated as release-blocking, consistent with the Quality Gate discipline already fixed in Appendix-M Section 25 — this checklist is the backend-specific instantiation of that same governing principle, completing the documentation chain from RGD v2.0 through to the operational backend codebase itself.